Assignment 4: RNN Deep Learning for Text Data & Sequences (BA_64061_001)

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Description and Goal

This assignment explores the use of Recurrent Neural Networks (RNNs) for analyzing text and sequence data through sentiment classification of movie reviews. Using the IMDB dataset, the objective is to evaluate how RNN architectures perform under different training conditions and embedding strategies. The task involves modifying the standard IMDB example by limiting reviews to 150 words, using the top 10,000 most frequent words, and training models on varying sample sizes while validating on a fixed set of 10,000 samples. Both an embedding layer and a pretrained word embedding are implemented before the Bidirectional layer to compare their effectiveness. The results aim to identify which approach yields better performance and how training sample size influences the benefits of using embeddings.

Dataset Summary

The dataset consists of 50,000 movie reviews used for sentiment classification. For analysis, only the 10,000 most frequent words are included in the vocabulary. The data is divided into multiple training subsets of different sizes (100, 1,000, 3,000, 5,000, 10,000, and 20,000 samples), while 10,000 samples are reserved for validation. Each review is truncated to a maximum length of 150 words to maintain consistency in input size.

Data Preprocessing

The dataset undergoes several preprocessing steps to prepare for model training. The training set size is restricted to specific sample counts to evaluate performance under varying data availability. Each review is truncated to a maximum of 150 words to maintain uniform input length. Additionally, only the 10,000 most frequent words are retained to limit the vocabulary and reduce model complexity.

Word Embedding

Word embeddings are used to represent words as numerical vectors in a high-dimensional space, allowing the model to capture semantic relationships between words more effectively than traditional one-hot encoding. Two embedding strategies are applied: a standard embedding layer trained within the model and pretrained embeddings such as GloVe, Word2Vec, or FastText. The pretrained embeddings provide rich semantic information learned from large text corpora, improving efficiency and potentially enhancing model performance without the need to train embeddings from scratch.

Model Architecture

The model is based on a Recurrent Neural Network (RNN) designed to handle sequential data, treating each movie review as a sequence of words. An embedding layer is incorporated to convert words into dense vector representations, enabling the model to capture semantic and contextual relationships within the text. This embedding layer can either use pretrained embeddings, such as GloVe, or be trained jointly with the model.

Key hyperparameters include the number of RNN units and the dimensionality of the embeddings, both of which determine the model's ability to represent and learn complex patterns in the data.

Training and Validation

The model is trained on the IMDB dataset using different training sample sizes: 100, 1,000, 3,000, 5,000, 10,000, and 20,000 reviews. A fixed validation set of 10,000 samples is used to evaluate performance consistently across experiments. Key metrics, including accuracy and loss, are recorded to compare the effectiveness of different embedding approaches and training sample sizes.

Methodology

1. Baseline Model

Model Description:
The baseline model employs a standard Recurrent Neural Network (RNN) combined with an embedding layer that learns word embeddings directly from the training data. This approach allows the model to capture sequential patterns in the text while constructing word representations tailored to the dataset.

Model Architecture:
The RNN processes the input text sequentially, taking one word at a time and updating its hidden states to capture contextual information. The embedding layer transforms each word into a dense vector, which serves as the input to the RNN. This setup allows the model to learn relationships between words and their influence on sentiment within a review.

Hyperparameters:
Key hyperparameters, such as the number of RNN units and the dimensionality of the embedding vectors, are defined and tuned to balance model complexity and learning capacity. These parameters influence how well the model can capture patterns and dependencies in the text.

Validation and Test Results:
The baseline model is evaluated on a fixed validation set of 10,000 samples. Validation

accuracy, test accuracy, and loss are recorded to establish a reference point for comparing other models and embedding strategies.

2. Model with Pretrained Word Embeddings

Utilization of Pretrained Word Embeddings:
Instead of learning embeddings from scratch, the model incorporates pretrained word embeddings such as GloVe. These embeddings provide semantic representations of words derived from large text corpora, allowing the model to leverage prior knowledge about word relationships.

Process of Loading Pretrained Embeddings:
Pretrained embeddings are loaded into the embedding layer of the RNN. Words in the dataset are mapped to their corresponding pretrained vectors, enabling the model to start training with embeddings that already capture semantic and contextual information. This can improve model performance, particularly when the available training data is limited.

Validation and Test Results:
The model with pretrained embeddings is evaluated using the same validation set. Metrics including validation accuracy, test accuracy, and loss are recorded to assess the performance gain obtained from leveraging pretrained semantic representations.

3. Varying Training Set Size

Training Set Size Adjustment:
To investigate the effect of data quantity on model performance, the size of the training set is systematically varied. Training subsets of 100, 500, 1,000, and 100,000 samples are used. This allows the analysis of how model performance scales with the availability of labeled data.

Validation and Test Results:
For each training subset, the model is trained and evaluated on the fixed validation set. Tracking performance across these varying training sizes provides insights into the relationship between dataset size and model generalization and highlights the point at which larger datasets improve performance significantly.

4. Comparison of Embedding Layer vs. Pretrained Embeddings

Performance Analysis:
A detailed comparison is conducted between models using an embedding layer trained from scratch and those utilizing pretrained word embeddings. The comparison is performed across

different training set sizes to understand how the choice of embedding strategy interacts with the volume of available data.

Model	Validation Accuracy(%)	Validation Loss	Test Accuracy(%)	Test Loss
One Hot Encoded model	80.1	0.46	79.2	0.463
Embedded Layer Model	75.4	0.512	74.7	0.522
Embedded Layer Model with Masking	79.1	0.454	79.2	0.453
Model with Pretrained GloVe Embeddings	78.4	0.454	78.7	0.449
Embedding Layer of 100 Training Samples	79.01	0.4569	79.08	0.4660
Pretrained Embedding Layer of 100 Training Samples	78.52	0.4582	77.88	0.4663
Embedding Layer of 1000 Training Samples	80.97	0.4263	78.06	0.4398
Pretrained Embedding Layer of 1000 Training Samples	79.56	0.4395	79.82	0.4402
Pretrained Embedding Layer of 3000 Training Samples	**81.53**	**0.4780**	**79.30**	**0.4905**
Embedding Layer of 3000 Training Samples	79.11	0.4546	**80.88**	0.4580
Embedding Layer of 5000 Training Samples	79.54	0.4611	78.64	0.4773
Pretrained Embedding Layer of 5000 Training Samples	78.85	0.4547	78.74	0.4574
Embedding Layer of 10000 Training Samples	81.12	0.4523	78.54	0.4789
Pretrained Embedding Layer of 10000 Training Samples	78.93	0.4512	79.84	0.4520
Embedding Layer of 20000 Training Samples	78.17	0.4602	78.79	0.4769
Pretrained Embedding Layer of 20000 Training Samples	78.67	0.4580	76.94	0.4588

Expected Outcome

The assignment was designed to compare how RNN models perform sentiment classification when using two different embedding strategies, a trainable embedding layer, and pretrained embeddings across varying training set sizes. The pretrained embeddings were expected to perform better with limited data since they've already learned from a large collection of texts. In contrast, the trainable embedding layer was expected to gradually improve as more data became available, allowing the model to learn domain-specific word relationships relevant to the IMDB reviews. Overall, it was anticipated that the embedding layer model would surpass the pretrained embedding model once the training set became sufficiently large.

Conclusion

The results show that the RNN model's performance depends strongly on both the embedding approach and the amount of training data. When the dataset was small (around 100 to 1,000 samples), the pretrained GloVe embeddings provided relatively stable results and slightly better performance due to their prior linguistic knowledge. However, as the number of training samples increased, the model with a trainable embedding layer began to outperform the pretrained model. From approximately 3,000 samples onward, the embedding layer model achieved higher validation and test accuracy across most runs.

Based on overall performance across all experiments, **the RNN model with a trainable embedding layer using 3,000 training samples** performed best. This configuration achieved the highest balance between accuracy and loss, demonstrating strong generalization without overfitting. It shows that while pretrained embeddings help in low-data situations, a trainable embedding layer yields superior results when sufficient data is available for the model to learn sentiment-specific representations.